

MedGemma recall benchmark: MDACE billing extraction

How far a 4B open model gets at pulling codable findings out of a clinical note.

EXECUTION PLAN
10 AUG 2026
SHIFAT HABIB

WHAT IS BEING MEASURED

MedGemma reads a note and extracts medical findings. **It does not produce codes.** Downstream, each extracted term goes into the SNOMED lookup, which returns the adjacent ICD code. Asking an LLM for codes directly compounds two hard problems into one and produces too many errors.

Extraction is therefore the ceiling on the whole pipeline: the lookup can never recover a code whose term was never extracted. This benchmark measures where that ceiling sits for **MedGemma-4B, zero-shot**. There is no baseline to beat. The output is a number, and Ehtesham sets the bar it has to clear.

DATA: ONE FILE, NOTHING ELSE

8-07-mdace-ner-eval_sample_100-LOCAL.jsonl . Self-contained: the note text is in the file, so no join and no sampling decisions.

24	100
NOTES	ANNOTATION ROWS
91	82
DISTINCT CODES	MODEL CALLS · ~1 H T4

Long notes are split into overlapping 400-word windows; extracted terms are pooled per note.

WHAT COUNTS AS A CORRECT EXTRACTION

Three columns describe each billed code. **Matching any one of them counts.**

Evidence text	what the note says (<i>depression</i>)
Code description	ICD catalogue wording (<i>Major depressive disorder, single episode, unspecified</i>)
SNOMED terms	clinical synonyms for the same concept

That gives a median of **4 accepted phrasings per row**, against 1 in the pilot. Recall is reported per row (100) and per distinct code (91).

HOW A MATCH IS DECIDED

Matching is approximate, not exact. Each level contains the one above it, and **recall is reported at every level** so the cost of each degree of looseness is visible rather than hidden in one threshold.

LEVEL	RULE	CATCHES
L1 exact	identical after normalising	baseline
L2 contains	one is a whole-word subset of the other	back pain ⊂ chronic back pain
L3 similar	word overlap + character similarity	typos, word order
L4 embeddings	biomedical sentence model, cosine	HTN → hypertension
L5 LLM judge	model decides if two phrases mean the same	deferred — see below

L4 is the level that earns its keep. Abbreviations are the largest single source of missed matches and no amount of string similarity reaches them: *CHF* and *congestive heart failure* share almost no characters. Only semantic matching closes that gap.

WHAT IS REPORTED

Recall	headline. Of the billed codes, how many were found
Precision, F1	secondary; false positives are watched, not ignored
Terms per note	the volume the recall was bought with
Not-in-note rate	hallucination: terms the model produced that aren't in the text

Every figure is given at every matching level.

THE NUMBER THAT MAKES RECALL MEANINGFUL

With loose matching and no volume control, a model that lists every phrase in the note scores near 1.00. Recall alone therefore cannot rank two models — a smaller, more verbose one would appear to beat a larger, more disciplined one.

So recall is always quoted **with the terms-per-note it was bought with**: “0.72 at 20 terms per note”, never a bare 0.72. This is what keeps the benchmark usable when we compare MedGemma against a larger model later.

PROMPT CHANGE

The pilot prompt asked the model to copy spans *verbatim*. That was right when the note's own wording was the only target; it is now wrong, because **HTN** can never string-match *Essential (primary) hypertension*.

The rewritten prompt asks for two fields per finding: the **span** as written in the note, and the **standard clinical name**. Either may match. MedGemma already knows the expansions, and the standard name is also what the SNOMED lookup wants downstream. To hold output length steady, the prompt asks for one entry per distinct finding and no repeats.

DEFERRED: THE LLM JUDGE

Recall rises at every level *by construction*, so L4 will always look best. The honest question is which level is still counting genuine matches, and the way to settle it is to have a model judge a sample of the pairs each level newly accepted.

It is slow and non-deterministic, so it is **not run for now**. It stays as the audit step, to be added when a specific level's number is challenged or when the result is close to the acceptance bar.

NEEDED FROM EHTESHAM

Full SNOMED lists	the file ships 24,333 of 270,287 concepts, or 9% . Every missing synonym is a match we cannot make. Procedure codes (CPT, ICD-10-PCS) have none at all
The lookup	running our extracted terms through the real SNOMED lookup replaces this proxy with the true end-to-end number, and ends the threshold discussion
Target recall	the figure that makes this a go

REFERENCE POINT

A 50-note pilot on the wider MDACE corpus, scored the strict way (exact matching, evidence text only) returned **recall 0.53**, against Ehtesham's independently observed ≈0.5. That is the floor this plan is designed to lift: every change above widens what counts as correct without touching the model.

Scope. Extraction only. ICD code prediction is a separate experiment and is not measured here.

MedGemma-4B-it · 4-bit · greedy · zero-shot